COMP 3500SEF · Final Report

Warehouse Logistics Management System
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Testing: Oscar, Li Mu Qiu.

1. Scope

Goals: traceable order decisions (each order's history auditable through ledger entries and notifications) and role-separated workflows. Roles: customer, staff, driver, administrator. Local-only: one app; seed data only (RK-10).

2. Development process

Increment closes on review or test.

Table A. Increments

Increment and date	Trigger	Outcome
Planning, 9–11 Oct 2026	Coursework kickoff	Scope, responsibilities and API contracts recorded (register)
17 Oct review	Stock and order review	Two defects; fixed as row 3
Confirmation, 26–28 Oct 2026	Row-2 defects: repeat submits, no ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Role functions integrated	All four role workflows (§1) run end to end
Driver and privacy, 2 Nov 2026	Driver tests outstanding	Both pass (privacy tests)
Diagram review, 10–11 Nov 2026	Diagrams diverged	Aligned (review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date export in three in-app interactions
CI health tests, 11–12 Nov 2026	No test covered CI-run outcomes	Two regression tests; suite reached 38 (unit.log)

3. Requirements

- FR-01 create orders with lines, totals Test. - FR-02 enforce lifecycle, stock release Test (cancel). - FR-03 number orders and shipments, track owners Test. - FR-04 inventory, ledger, low-stock alerts Test. - FR-05 record, read notifications Test. - FR-06 admin aggregates, CSV Test. - FR-07 authenticate, access Test. - FR-08 assign orders, record delivery Test.

- NFR-01 performance: API p95 ≤2 s, first contentful paint ≤3 s (first content shown) (measurement). - NFR-02 CI health — every commit ends in green unit, integration, browser runs (runs, observations). - NFR-03 least privilege: role gates; delivery address only to the assigned driver (privacy tests). - NFR-04 interaction budget: every core operation (the eleven in §8) takes at most three in-app interactions; one is a fill, selection or click (walkthrough). - NFR-05 process quality — every increment closes on a recorded review or test run.

Table B. Risk coverage

Risk	Addressed by	Evidence
①RK-01, RK-03, RK-06, RK-08 schedule, test week, coursework, video	weekly re-planning, scope priorities	risk plan
②RK-02 contract rework	lifecycle locked as test constants	constants tests
③RK-04, RK-07 setup, stack	setup guide; CI per commit	README ci.yml
④RK-05 demo data	deterministic seed, regression-tested	seed test

Risk	Addressed by	Evidence
⑤RK-09, RK-10 hosting; personal data	accepted — out of scope (§1); seed data only	risk plan

4. System models

Layers: pages, routes, handlers, Prisma/SQLite.

Order items reference one product; orders belong to one user; ≤1 shipment records events; products accumulate movements (InventoryLog); notifications relate to orders.

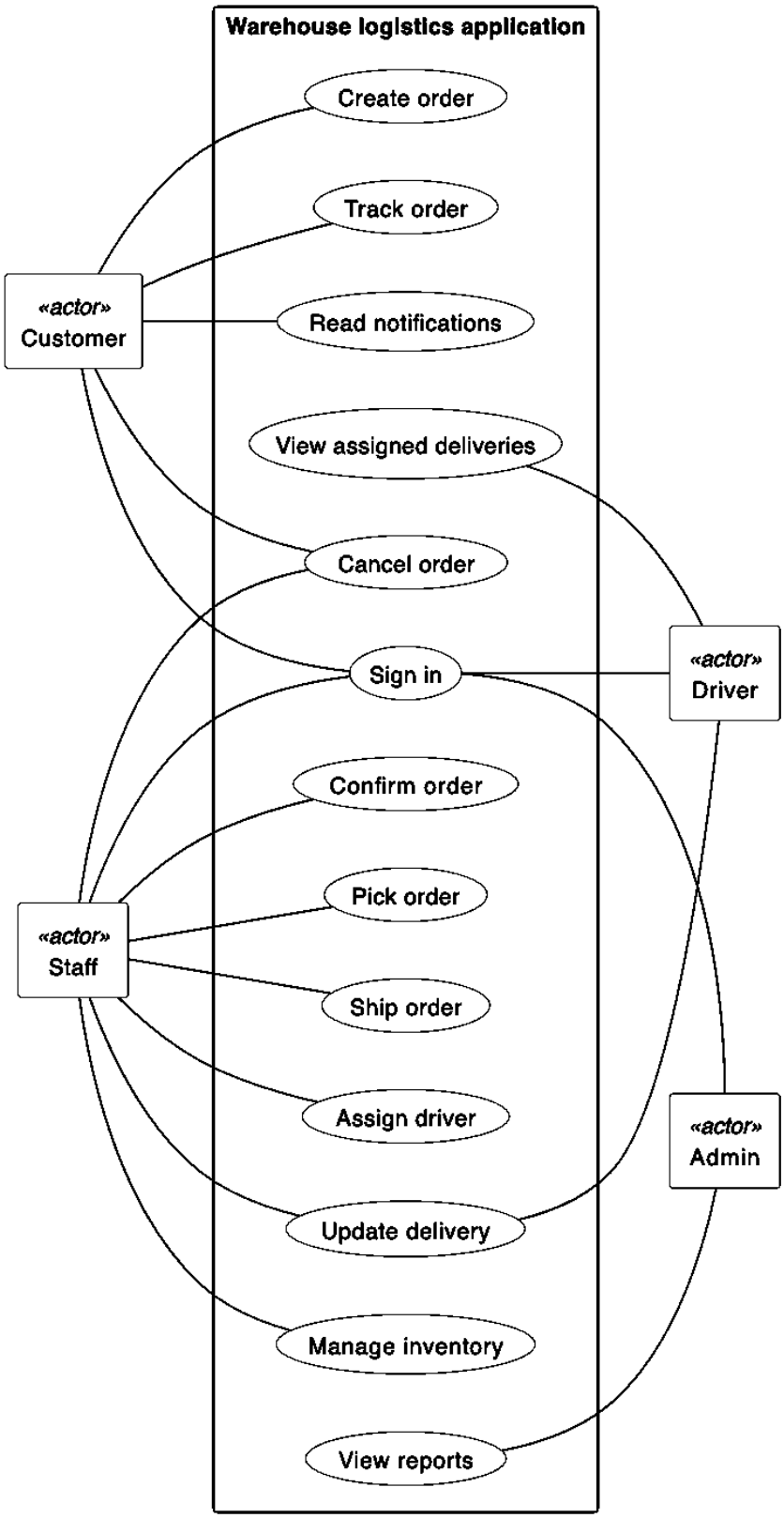


Figure 1. Thirteen use cases.

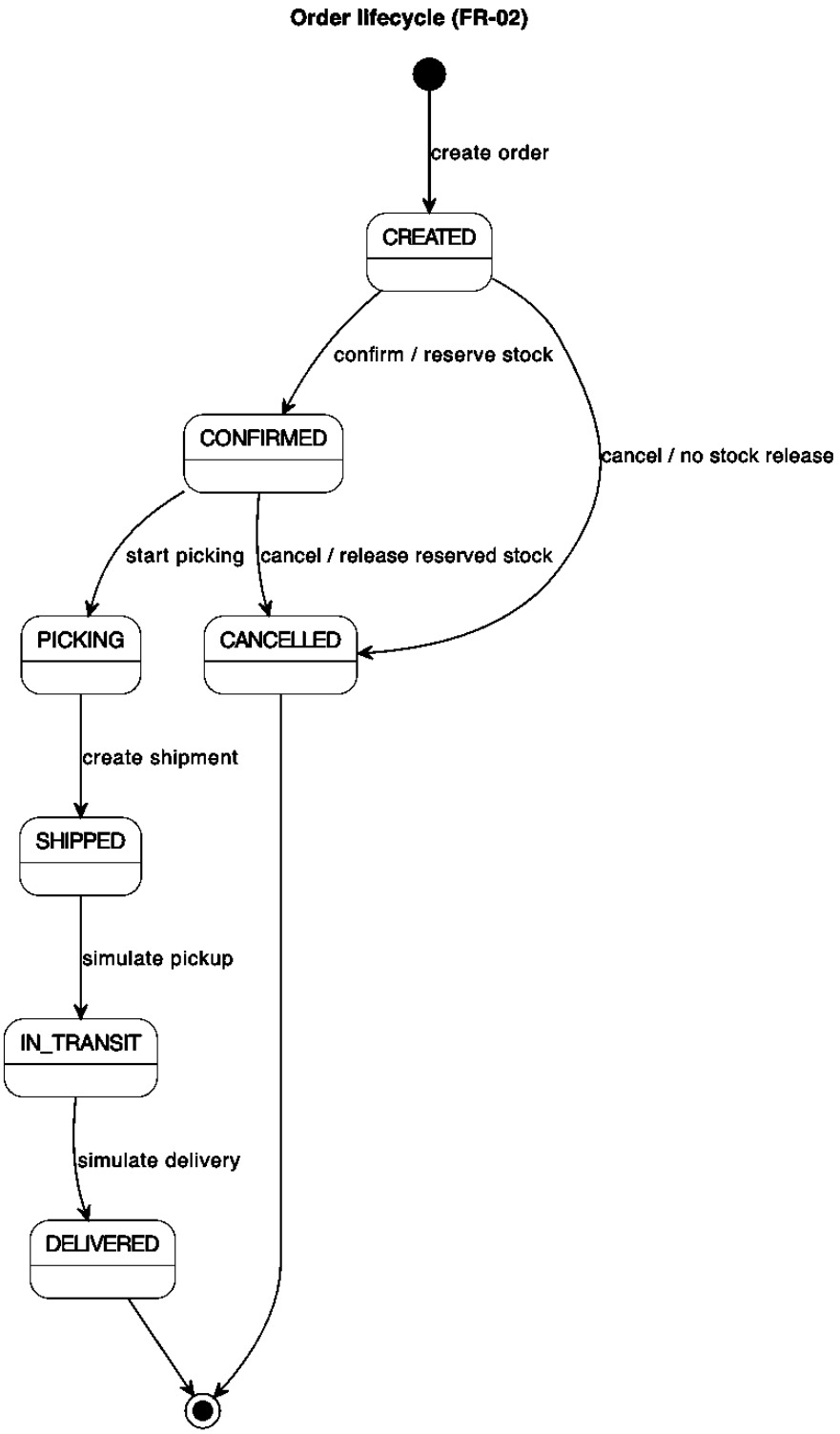


Figure 2. Order transitions and stock effects: reserve, release, or unchanged.

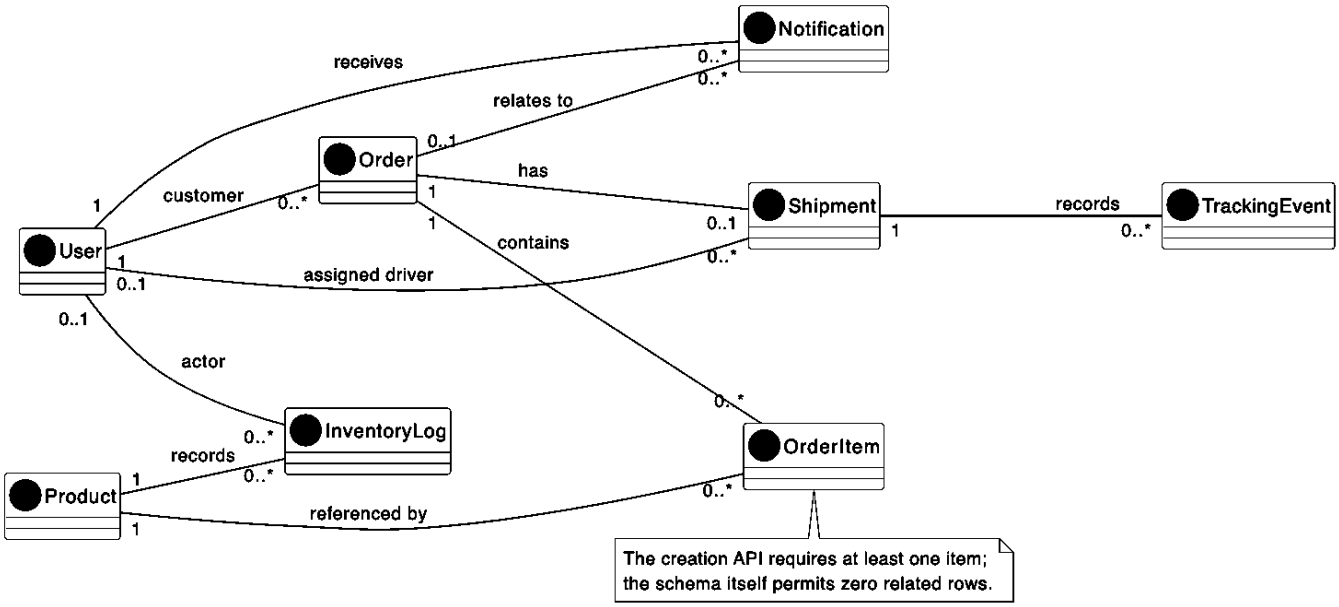


Figure 3. Entities and multiplicities.

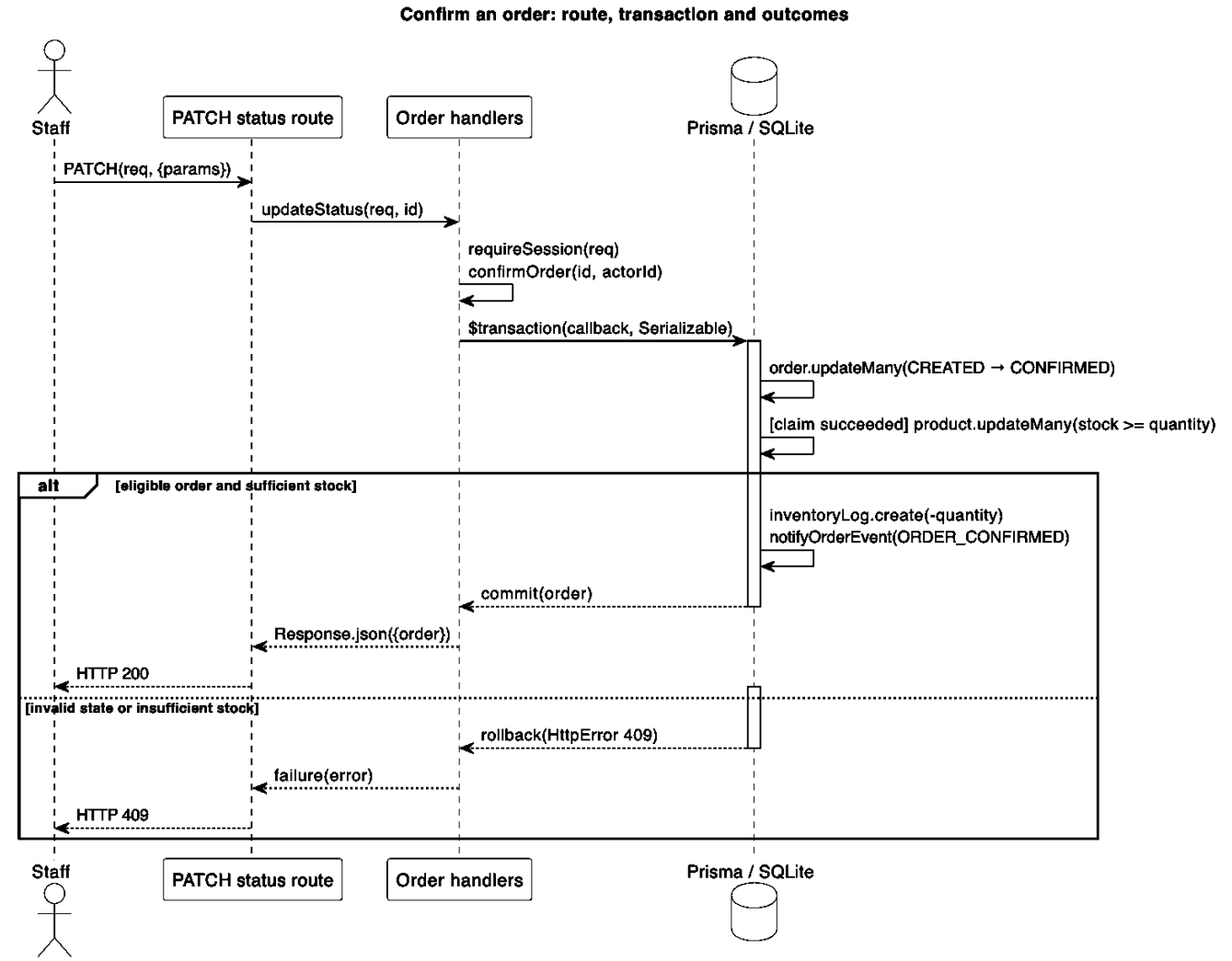


Figure 4. Confirmation: guard, writes, rollback.

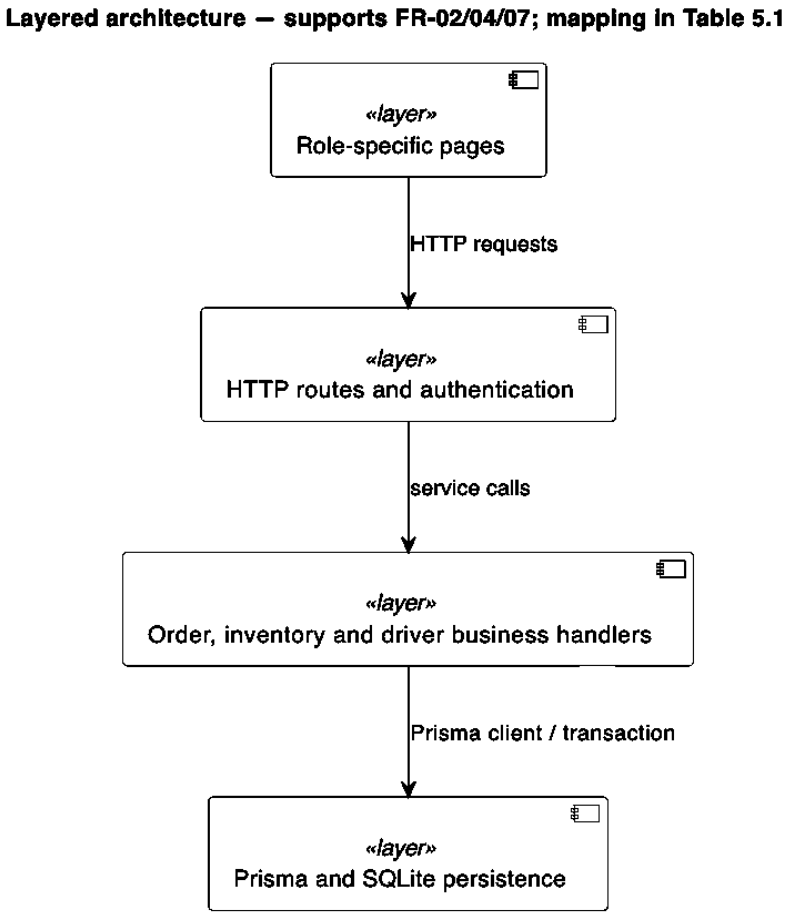


Figure 5. Four layers.

5. Implementation

Confirmation reserves stock; lifecycle CREATED→CONFIRMED→PICKING→SHIPPED→IN_TRANSIT→DELIVERED; CANCELLED from CREATED or CONFIRMED.

Role	Transition and effect
Customer	CREATED: cancel, no restock
Customer	CONFIRMED: cancel, restock
Staff	CREATED: rejected
Staff	CONFIRMED: cancel, restock
Customer, staff	PICKING onwards: rejected

The lifecycle is locked as test constants (tests); staff have no refuse action by design.

Confirmation is atomic (409 on repeats; rollback on short stock); driverId set at SHIPPED, kept; FR-03/04 verified (order-api, driver-api, inventory-api).

6. Verification

Verification checks the implementation against §3; validation exercises the delivered workflows (§8). CI records 38 unit, 140 integration, 2 integrity checks: the demo-data reset requires RESET_DEMO_DATA=YES, and the bundle's SHA-256 manifest must match the source (integrity checks). Logs: integration "tests 140"; unit "Tests 38 passed (38)".

Loopback measurement: p95 3.2–13.6 ms, first paint 64–96 ms vs 2/3 s.

The three reviews (Table A rows 2, 6, 7): rows 2 and 6 verified requirements and models; row 7's fix was validated through the interface (§8).

7. Evolution

Three reviews: 17 Oct found two defects (repeat submits and a missing ledger) fixed by §5; 10–11 Nov aligned diagrams, traceability, rewrote NFR-01/02, the window becoming CI-run checks; 11 Nov added the direct export path (issue #15). Feedback that the CI record was untested added two regression tests (row 8; suite 36→38).

8. Validation

Walkthrough covers eleven operations (of Figure 1's thirteen): create order; open details; confirm; start picking; ship; assign driver; record pickup; record delivery; filter report; export current; export selected dates; at most three interactions each. "Browser acceptance passed: 26 checks": 20 operation checks (five operations once, three twice, three thrice) plus 6 cross-cutting (two each: validation, session, interface). Records from Chung Ngon Man and Li Mu Qiu: 64/64 task attempts completed unaided across eight increments; median 2.1 min (evidence/user-validation.md).

9. Feedback and reconciliation

Sources: whole-team reviews; test runs and the scripted walkthrough executed by Oscar and Li (Testing).

Table C. Feedback

Feedback (source, date)	Disposition and evidence
Diagrams diverged (10–11 Nov, corrections)	Corrected
Stock (FR-04) (17 Oct, traceability)	Moved to FR-02
Thin triggers (10–11 Nov, register)	Rewritten
Filter-then-export: four interactions (11 Nov, issue #15)	Combined path rejected — its steps already totalled four; the direct path meets NFR-04; issue closed
NFR-01/02 assumed hosted measurement and a daily window (10–11 Nov, revision log)	NFR-02: CI-run checks; NFR-01: loopback (§6)

10. Evidence

38 unit, 140 integration, 2 integrity (12 Nov; tests only since the browser run), 26 browser checks (11 Nov) — BASELINE. Traceability.